

Problem 1. Fibonacci Divisibility

(a)

Proof by strong induction. Let $P(n) := F_{n+a} \equiv F_n \cdot F_{a+1} \pmod{F_a}$, now to prove $p(n)$ for $n \geq 0$ by strong induction.

Base Case:

$$P(0) = F_a \equiv F_0 \cdot F_{a+1} \equiv 0 \pmod{F_a}.$$

$$P(1) = F_{a+1} \equiv F_1 \cdot F_{a+1} \equiv 1 \cdot F_{a+1} \equiv F_{a+1} \pmod{F_a}$$

Therefore, both cases hold.

Inductive step: For strong induction on n assume $P(n-1)$ and $P(n-2)$ holds for $n \geq 2$. So,

$$F_{(n-1)+a} \equiv F_{n-1} \cdot F_{a+1} \pmod{F_a} \quad (1)$$

$$F_{(n-2)+a} \equiv F_{n-2} \cdot F_{a+1} \pmod{F_a} \quad (2)$$

Now by property of the congruence on equation (1) and (2),

$$F_{(n-1)+a} + F_{(n-2)+a} \equiv F_{n-1} \cdot F_{a+1} + F_{n-2} \cdot F_{a+1} \pmod{F_a}$$

$$F_{n+a} \equiv (F_{n-1} + F_{n-2}) \cdot F_{a+1} \pmod{F_a}$$

$$F_{n+a} \equiv F_n \cdot F_{a+1} \pmod{F_a}$$

Which implies $P(n)$ is true. Therefore by strong induction hypothesis $P(n)$ is true for all $n \geq 0$. $\square$

(b)

Proof by induction. Induction hypothesis Assume $P(n)$ is true, i.e.,

$$F_{na} \equiv 0 \pmod{F_a}.$$

Base Case: $P(0) = F_0 \equiv 0 \pmod{F_a}$

Base case holds. We now prove $P(n+1)$, namely,

$$F_{(n+1)a} \equiv 0 \pmod{F_a}.$$

$$F_{na+a} \equiv 0 \pmod{F_a}.$$

Using part (a) with n replaced by na ,

$$F_{na+a} \equiv F_{na} F_{a+1} \pmod{F_a}.$$

$$F_{na+a} \equiv 0 \cdot F_{a+1} \pmod{F_a}.$$

$$F_{na+a} \equiv 0 \pmod{F_a}.$$

Which proves $P(n+1)$ holds and by induction hypothesis $P(n)$ is true for $n \geq 0$. $\square$

Now if $a|b$ then $\exists k \in \mathbb{N}$ such that $a * k = b$. Now by $P(k)$, $F_{ak} = F_b \equiv 0 \pmod{F_a}$ which implies $F_a|F_b$.